

P-Hacking

How Data Can Lie

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“Chocolate helps you lose weight.”

“Coffee causes cancer.”

“Eating ice cream extends your life.”

**Why do we often see headlines that
contradict each other?**



P-Hacking

The practice of manipulating analysis or research methods to get significant results.

- Inflate the significance of findings
- Reporting results selectively
- Running many tests until a <0.05 p-value is obtained

What Impacts Does P-Hacking Have?

Impacts many industries including:

- **Social Science:** Psychology, Sociology, Economics
- **Health Science:** Drug Clinical Trials, Public Health, Epidemiology
- **Technology:** Computer Science, Cybersecurity, Finance

Leads to false beliefs, misleading conclusions, poor policy decisions, other consequences.

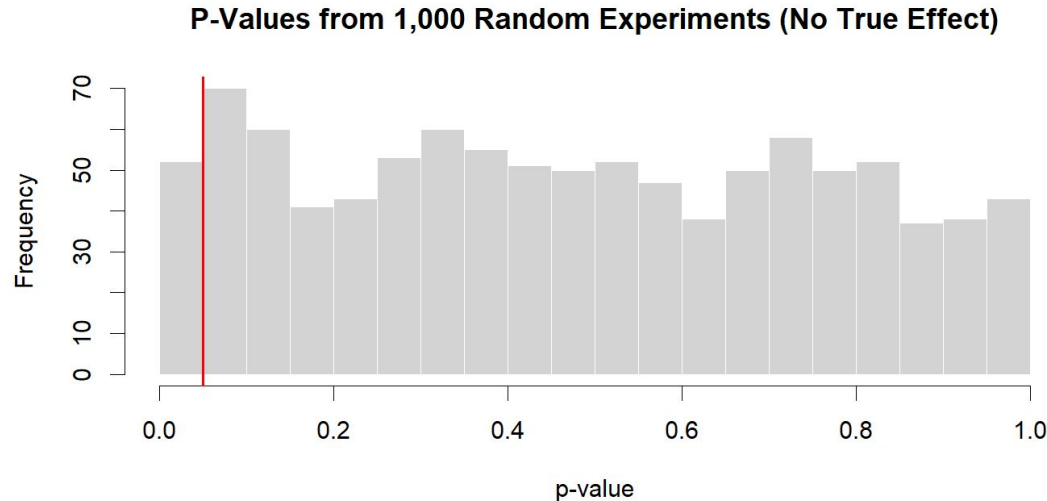
Why? “Publish or perish” culture in research.

Statistical Understanding



What is a P-Value?

- The probability of observing results this extreme if the null hypothesis is true
- $p < .05$ means results this extreme would occur less than 5% of the time by chance alone



Multiple Testing Problem

If a 0.05 p-value means that we will get a false positive 5% of the time, then what will happen if we perform 20 tests?

$$1 - (0.95)^{20} \approx 0.642$$

If we perform 20 tests, our chances of getting at least one false positive rises to **64.2%**.

The probability continue to rise as more tests are done.

Methods of P-Hacking

- Cherry Picking
 - Selectively choosing data that supports the desired conclusion
 - Excluding data points that weaken significance
- Data Dredging
 - Testing many variables, subgroups, or relationships, “fishing” for significance
- HARKing (Hypothesizing After Results are Known)
 - Rewriting a hypothesis after seeing the data
 - Presenting findings as though they were predicted from the beginning

R Demonstration



Cases of P-Hacking



The “Chocolate Weight Loss” Study (2015)

What happened:

A study claimed that eating chocolate every day helps you lose weight. They even invented a fake research institute (“Institute of Diet and Health”). It went viral worldwide.

How p-hacking happened:

- Researchers measured 18+ variables (weight, sleep, cholesterol, sodium, mood, etc.).
- With such a small sample size, one variable was bound to look “significant” by chance.
- They published only the “positive” result and ignored the rest.

This shows how easy it is to produce “significant” results with bad methods — and how quickly the media spreads them.

“Power Posing” (2010)

What happened:

A famous psychology study claimed that standing in “power poses” boosts hormones and confidence.

How p-hacking happened:

- Many different hormone and behavioral outcomes were measured.
- Only the “significant” ones were reported.
- Later, large replication attempts **failed** to reproduce the findings.

This was put into a TED Talk with millions of views before the science was questioned

Debt percentage of GDP affects (2010)

What Happened:

- Paper claimed countries with debt $> 90\%$ of GDP experience **sharp drops in economic growth**.
- Became one of the most influential economics findings of the decade.
- Used to justify austerity policies in the U.S. and Europe after the 2008 recession.
- A grad student (Herndon, 2013) attempted replication and uncovered major errors.

How p-hacking happened:

- Selective Data Exclusion: Omitted several high-debt countries that had *strong* growth (e.g., Australia, Canada, New Zealand).
- Arbitrary Weighting Choices: Countries with 1 year of data were weighted the same as those with 19 years — inflating the negative effect.
- Lack of Transparency: Data and code were not shared, preventing replication for years.

How to Spot P-Hacking



How to know if it might be p-hacking:

What do the examples have in common?

- Too many variables measured but only a few reported
- Pressure to publish exciting findings

Things to look out for

- P-values cluster just below 0.05
- No preregistration or analysis plan
- Lack of replication or failure to replicate
- Inconsistent sample sizes or impossible statistics

How to Prevent P-Hacking

- Preregister hypotheses and analysis plans
 - Commit to outcomes, models, and exclusions before seeing the data
- Use a fixed sample size (no optional stopping)
- Correct for multiple comparisons (bonferroni?)
- Report all analyses, not just significant ones
- Distinguish between exploratory and confirmatory analysis
 - Exploratory is fine but don't state it like a fact when data doesn't show for it.